



Quantum spin ice in three-dimensional Rydberg atom arrays

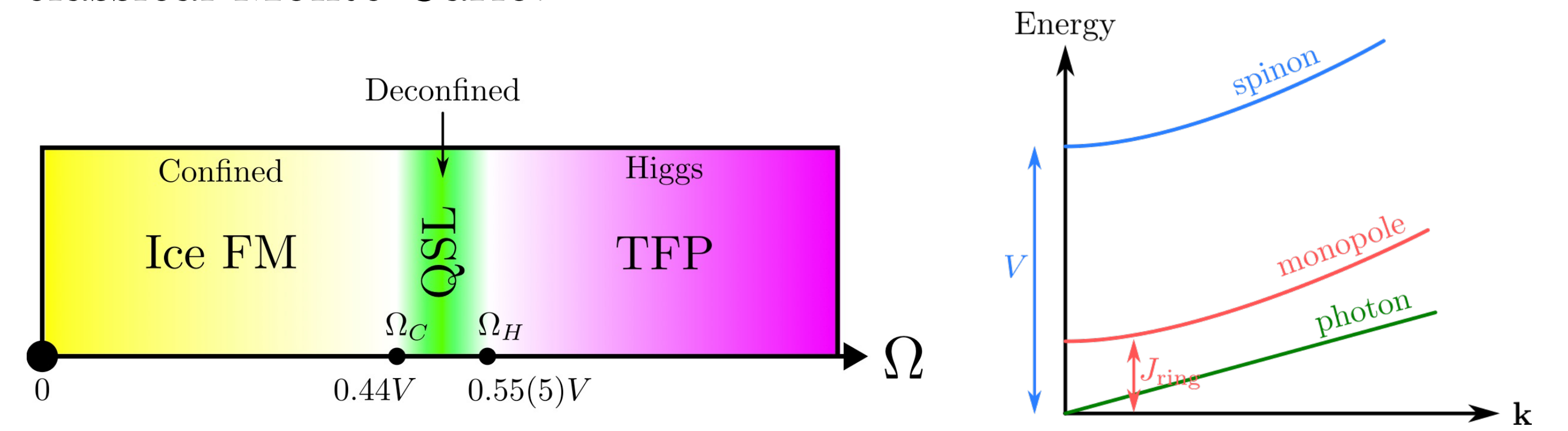
Jeet Shah, Gautam Nambiar, Alexey V. Gorshkov, and Victor Galitski

Background & Motivation

- Quantum spin liquids (QSLs) are states which are disordered because of the quantum fluctuations and are highly entangled.
- They are typically described by the deconfined phases of gauge theories and exhibit fractionalization.
- They cannot be distinguished from ordered states by any local order parameter. Non-local order parameters are necessarily required which are difficult to measure in traditional condensed matter systems.
- Recently, signatures of a $\mathbb{Z}_2$ QSL were observed in a 2D Rydberg atom array (Semeghini et al.) and the toric code ground state was prepared at Google (Satzinger et al.).
- However, there is no conclusive evidence of a $U(1)$ QSL in any system so far, which is the focus of our work.

Phase diagram

- The phase diagram is determined by calculating $\langle \hat{H}_{\text{eff}} \rangle$ in $|\psi_{IFM}\rangle$, ordered and $|\psi_{RK}\rangle$, QSL phases. The latter is calculated using classical Monte Carlo.

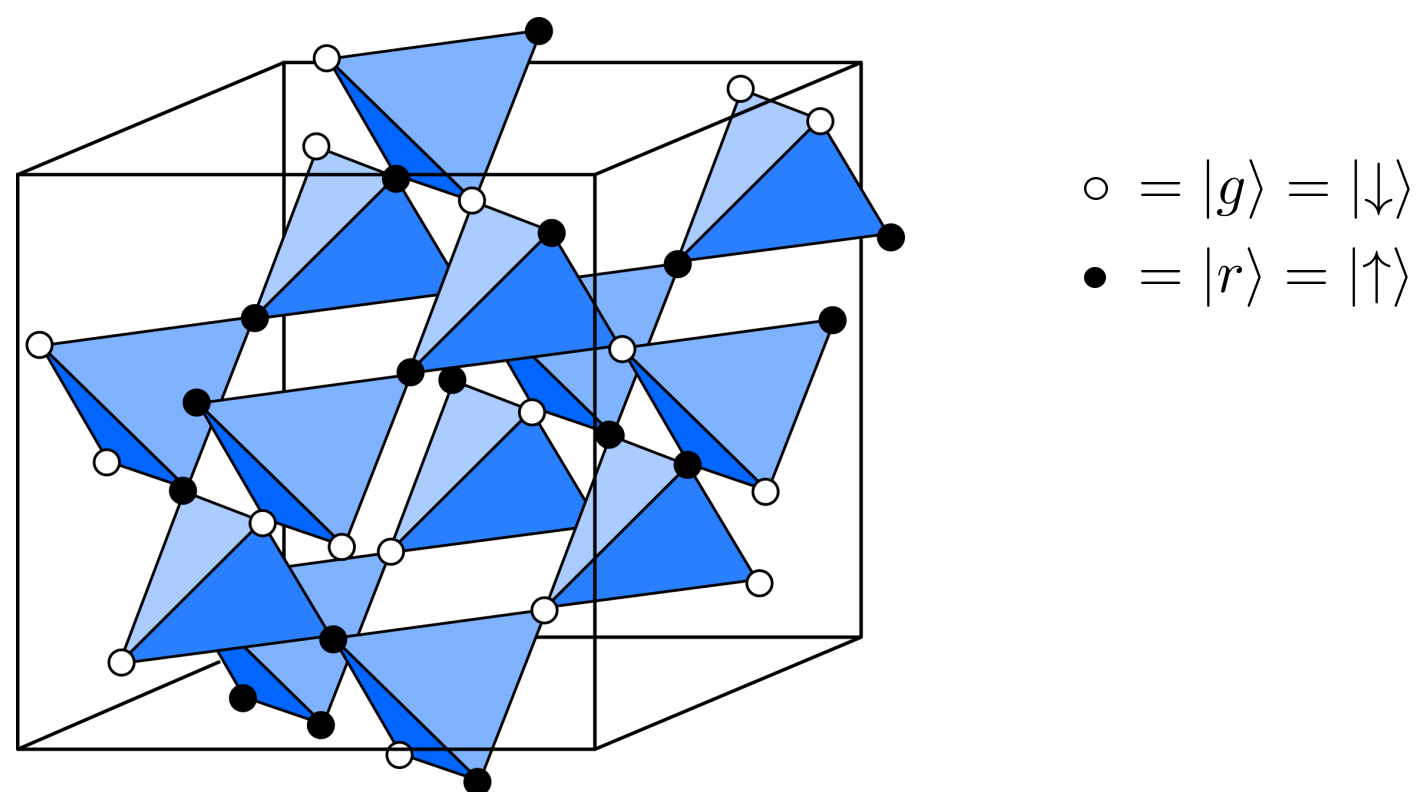


- A QSL state can be prepared in two ways: (1) Adiabatic preparation where one starts with $\delta \ll -V$ and $\Omega \ll V$ and tunes δ and Ω adiabatically to the desired region. (2) Start with $\delta \ll -V$, $\Omega \ll V$ and evolve the system adiabatically w.r.t the spinons but suddenly w.r.t the monopoles. This projects out the states outside the ice manifold and prepares the RK wavefunction.

The model

- Consider neutral atoms trapped on the sites of a pyrochlore lattice using optical tweezers. The Hamiltonian is given by

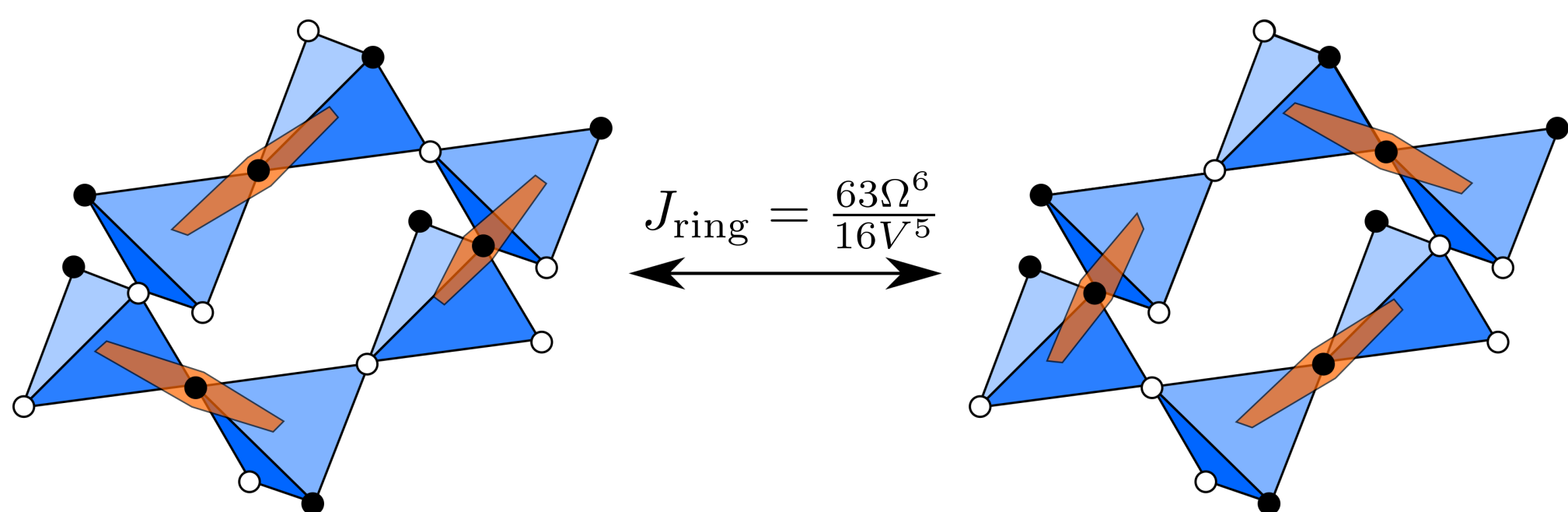
$$\hat{H}_{\text{ryd}} = \underbrace{-\delta \sum_i \hat{n}_i}_{\text{Laser detuning}} + \underbrace{\frac{V}{2} \sum_{i \neq j} \left(\frac{a}{|\mathbf{x}_i - \mathbf{x}_j|} \right)^6 \hat{n}_i \hat{n}_j}_{\text{Van der Waals Interaction}} + \underbrace{\frac{\Omega}{2} \sum_i (\hat{b}_i + \hat{b}_i^\dagger)}_{\text{Rabi drive}}$$



- First, we assume that $\Omega \ll V, \delta$ and ignore the long-range interactions. Their effects will be considered later. The Hamiltonian can now be written as

$$\hat{H} = \underbrace{\frac{V}{2} \sum_{\text{tet}} (\hat{n}_{\text{tet}} - \rho)^2}_{\hat{H}_0} + \underbrace{\frac{\Omega}{2} \sum_i (\hat{b}_i + \hat{b}_i^\dagger)}_{\hat{H}_\Omega}$$

- Treating $\hat{H}_\Omega$ as a perturbation over $\hat{H}_0$ gives ring exchange terms at sixth order. By adding a Rokhsar-Kivelson (RK) potential term, one can go to the RK point near which the system is a $U(1)$ QSL^[1].



- When the long-range interactions are considered, we show that the ground state is a $\mathbf{0}$ momentum state in the ice manifold when $\Omega = 0$. We call it the ice ferromagnet (ice FM) state, $|\psi_{IFM}\rangle$.
- For $\Omega \neq 0$, we treat both the long-range interactions and the Rabi drive upto sixth order in perturbation theory and obtain the effective Hamiltonian, $\hat{H}_{\text{eff}}$.

Diagnosis of the QSL

- Following correlators can distinguish between the phases.
- Plaquette-plaquette correlators: These tell if the photon is gapless. The correlators are, $\langle \hat{X}_P \hat{X}_{P'} \rangle$ and $\langle \hat{Y}_P \hat{Y}_{P'} \rangle$, where

$$\hat{X}_P = (\hat{S}_1^+ \hat{S}_2^- \hat{S}_3^+ \hat{S}_4^- \hat{S}_5^+ \hat{S}_6^- + \text{H.c.})$$

$$\hat{Y}_P = -i \hat{S}_1^+ \hat{S}_2^- \hat{S}_3^+ \hat{S}_4^- \hat{S}_5^+ \hat{S}_6^- + \text{H.c.}$$

- Monopole correlator shown below, decays exponentially in the QSL phase and is a nonzero constant in the confined phase.

$$\langle \hat{\mathcal{M}}^\dagger \hat{\mathcal{M}}(\mathbf{r}_1 \rightarrow \mathbf{r}_2) \rangle = \langle e^{i \sum_{\mathbf{x} \in \text{pyrochlore}} \mathcal{A}_{\mathbf{x}} (\hat{n}_{\mathbf{x}} - 1/2)} \rangle$$

- Bricmont-Fröhlich-Fredenhagen-Marcu (BFFM) order parameters shown below are exponentially small in the QSL while are nonzero constants in the ordered phases.

$$\chi_C^E = \frac{|\langle e^{i \sum_C \hat{a}_{\mathbf{r}\mathbf{r}'} + \text{H.c.}} \rangle|}{\sqrt{|\langle e^{i \sum_C \hat{a}_{\mathbf{r}\mathbf{r}'} + \text{H.c.}} \rangle|}}, \quad \chi_C^M = \frac{\langle \hat{\mathcal{M}}^\dagger \hat{\mathcal{M}}(\mathbf{r}_1 \xrightarrow{C} \mathbf{r}_2) \rangle}{\sqrt{\langle \hat{\mathcal{M}}^\dagger \hat{\mathcal{M}}(\mathbf{r}_1 \xrightarrow{C} \mathbf{r}_1) \rangle}}$$

Results

- We find that the system is in a $U(1)$ QSL phase in a window of detunings and Rabi frequencies.
- The Higgs transition and the confinement-deconfinement transition can be accessed by varying the Rabi frequency.
- We comment on the state preparation and propose various correlators including non-local correlators that can distinguish the phases. We also provide protocols to measure them.

References

- M. Hermele, M. P. A. Fisher, and L. Balents, Phys. Rev. B **69**, 064404 (2004).
- N. Shannon, O. Sikora, F. Pollmann, K. Penc, and P. Fulde, Phys. Rev. Lett. **108**, 067204 (2012).
- R. Verresen, M. D. Lukin, and A. Vishwanath, Phys. Rev. X **11**, 031005 (2021).